

When the Lens Goes Blind

The phase cycle of the Li-Sekatskii detector, and the price of a shift taken too far

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Summary

A companion note, [li_lens_law.md](#), found that the shift in the **Li-Sekatskii coefficients** behaves as a lens with an exact optimal width: for a hypothetical zero at $\rho = 1/2 + \beta + iT$, writing $d = 1/2 - a$ and $R = |\rho - 1/2|$, the exponential rate $r(d) = \log|q_a(\rho)|$ is maximal at $d = R$.

That is a statement about the **modulus** of a complex number. This note takes the **phase**, and then follows the rate to the far end where **Sekatskii's Theorem 3** lives.

Three things come out of the phase, and two of them are negative.

The rate has a closed form and a symmetry.

$$\tanh r(d) = 2\beta d / (d^2 + \beta^2 + T^2),$$

which is invariant under $d \mapsto R^2/d$. So every sub-maximal rate is attained at *two* lens widths whose product is R^2 , and $d = R$ is that map's only fixed point — the optimum is a centre of symmetry, not merely a stationary point. In the coordinate $x = \log(d/R)$ the rate is $\operatorname{artanh}((\beta/R)\operatorname{sech}x)$: a **sech dome on the logarithmic axis**, so the natural distance from the optimum is the ratio d/R , not $d - R$.

At the optimum the phase is exactly $-\pi/2$, independent of β and T . The quartet's contribution therefore runs on a strict cycle of period four — strong detection at $n \equiv 0(\bmod 4)$, strong **masking** at $n \equiv 2$, and no exponential term at all for odd n . **In 24 measured runs the first negative index of the full coefficient was $\equiv 0(\bmod 4)$ every time.**

And the far end is Sekatskii's. As $d \rightarrow \infty$ the image $q_a(\rho) \rightarrow -1$, every C_n turns positive, and the detection index grows **linearly** in the lens width. That is his Theorem 3 seen from the rate curve: $r(d)$ vanishes at both ends and peaks at R , so a shift taken far past the optimum does not merely fail to help — it hides. Temporarily: for any fixed shift the rate stays positive and the violation returns.

The two negative results. Lenses tuned to align the phase at a chosen order do **not** improve the first detection index; and the far detection pockets, though real, are swamped by the background by seven orders of magnitude. Both look promising and neither works, which is why they are here.

Status legend

tag	meaning
[K]	Known. Source named in §1. No novelty claimed.
[P]	Proved here, and machine-checked step by step.
[M]	Measured, with parameters stated.
[C]	Caveat.
[O]	Open.

1. Prior art, and what this note is not

component	where it lives
the shifted map, the derivative object, and the equality between them	the Li-Sekatskii coefficients — Sekatskii, arXiv:1304.7895 = <i>Ukrainian Math. J.</i> 66 (2014) 415–431

component	where it lives
positivity of $\lambda_n \iff \text{RH}$, and the multiset theorem behind it	Li (1997); Bombieri & Lagarias (1999)
the masking statement, proved	Sekatskii , <i>On the Generalized Li's Criterion...</i> , Springer Proc. Math. Stat. 348 (2021) 241–254, Theorem 3 : for any m there is c such that $\lambda_{n,b,1/2} = \sum_{\rho} (1 - ((\rho+b)/(\rho-b-1))^n) > 0$ for all $n \leq m$ and all $b \geq c$ the companion note, li_lens_low.md
the exact optimum $d = R$ and the factor- T gain	
explicit detection indices for Li-type coefficients	Brown (2005); Bucur , Ernvall-Hytönen , Odžak & Smajlović (2016); Palojärvi (2019)

[C] The relation to Theorem 3 must be stated carefully, because it is easy to overstate. Theorem 3 is proved for ξ itself, and Sekatskii confirms (correspondence, August 2026) that it is obtained through the analytic formula for ξ'/ξ — so the proof uses what ξ is, not only the shape of the sum. What he describes as *apparently* claimable is the corresponding statement for general multisets containing off-line elements, which is a separate matter. **Everything measured below is the multiset case** — a quartet planted at $1/2 \pm \beta \pm iT$ among genuine critical-line ordinates — so it bears on that extension, not on the theorem as proved. With $b = d - 1/2$ his $\lambda_{n,b,1/2}$ is exactly the sum computed here.

What is offered as new

1. **[P]** the closed form $\text{tanh}r(d) = 2\beta d/(d^2 + \beta^2 + T^2)$, its exact invariance under $d \mapsto R^2/d$, and the sech profile on the logarithmic axis;
2. **[P]** the phase identity $\varphi(R) = -\pi/2$ and the resulting $n \bmod 4$ cycle;
3. **[M]** that the cycle governs the **full** coefficient, not only the isolated quartet — 24 of 24 first-negative indices $\equiv 0 \pmod{4}$;
4. **[M]** the scaling $N_M(d) \cdot r(d) = \log M + O(1)$ on 100,000 ordinates, the U shape with its minimum at $d = R$, and linear growth of the index in the lens width;
5. **[M]** two negative results, §6.

2. The rate, closed

Write $\rho = 1/2 + \beta + iT$ with $\beta > 0$, $d = 1/2 - a$, and $R = \sqrt{(\beta^2 + T^2)} = |\rho - 1/2|$.

Proposition 1 [P]. $\text{tanh}r(d) = 2\beta d/(d^2 + \beta^2 + T^2)$.

Proof. $e^{2r} = N/D$ with $N = (\beta + d)^2 + T^2$ and $D = (\beta - d)^2 + T^2$, so $\text{tanh}r = (N - D)/(N + D)$. Now $N - D = 4\beta d$ and $N + D = 2(\beta^2 + d^2 + T^2)$. ■

[C] The $\beta \ll T$ form is different and is not the identity. Dropping β^2 gives $2\beta d/(d^2 + T^2)$, which agrees only to the order shown: at $d = 30$, $\beta = 0.1$, $T = 30$, the exact value is 0.0033333148 against 0.003333333.

Proposition 2 [P]. The rate is invariant under $d \mapsto R^2/d$, whose only fixed point in $d > 0$ is $d = R$.

Proof. Substituting R^2/d into $2\beta d/(d^2 + R^2)$ returns it unchanged; $d = R^2/d$ gives $d = R$. ■

So the sensitivity landscape is symmetric about the optimum **multiplicatively**. A lens k times too narrow and one k times too wide have the *same* rate.

Corollary 3 [P]. With $x = \log(d/R)$,

$$r(x) = \operatorname{artanh}(\beta/R) \cdot \operatorname{sech} x,$$

even in x , peaked at $x = 0$. **The peak is broad on the ratio scale:** for $\beta \ll T$ one keeps half the maximal rate over $0.268R \lesssim d \lesssim 3.73R$. Matching the lens matters; matching it precisely does not.

3. The phase

3.1 The value at the optimum

Rationalising, $q_a(\rho) = [(R^2 - d^2) - 2idT]/[(\beta - d)^2 + T^2]$.

Proposition 4 [P]. At $d = R$ the real part of the numerator vanishes identically, so $q_a(\rho) = -i|q_a(\rho)|$ and

$$\varphi(R) = -\pi/2,$$

independent of β and T .

[M] Checked at $(\beta, T) = (0.1, 30), (0.01, 500), (0.4, 10), (0.001, 3)$: the computed phase equals $-\pi/2$ to 10^{-14} in every case.

3.2 The quartet, and the cycle of four

The symmetric quartet $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$ maps to $e^{\pm r \pm i\varphi}$, and since $q_a(1 - \rho) = 1/q_a(\rho)$,

Proposition 5 [P]. $C_n(d) = 4 - 4\cosh(nr(d))\cos(n\varphi(d))$.

Checked against the direct sum over the four zeros to 5×10^{-14} .

At $d = R$ this becomes $4 - 4\cosh(nr)\cos(n\pi/2)$, and the cycle is exact:

$n \bmod 4$	$\cos(n\varphi)$	$C_n(R)$	
0	+1	$4 - 4\cosh(nr) < 0$	detection
1	0	$= 4$ exactly	no exponential term
2	-1	$4 + 4\cosh(nr) > 0$	masking
3	0	$= 4$ exactly	no exponential term

[M] At $\beta = 0.1$, $T = 30$: $C_{100}(R) = -0.224287$, $C_{102}(R) = +8.233435$, and $C_{101}(R) = C_{103}(R) = 4.000000$.

So the lens that maximises the envelope fails at three orders in four, in two different ways. At $n \equiv 2(\text{mod}4)$ — one order in four — the off-line quartet contributes a large *positive* amount and actively conceals itself, which is worse than useless. At odd n — half the orders — its exponential growth simply does not enter the real part, so the quartet is invisible rather than misleading. Only $n \equiv 0(\text{mod}4)$ detects.

3.3 The cycle survives into the full coefficient

The above is arithmetic on four planted points. The question is whether it governs $S_n = \Sigma_\rho(1 - q_a(\rho)^n)$ with real ordinates present.

[M] It does. Over 24 runs — six heights $T = 10, 20, 30, 50, 80, 120$, two displacements $\beta = 0.1, 0.05$, and two ordinate counts 2000, 8000 — the first index at which the full coefficient goes negative was $\equiv 0(\text{mod}4)$ in **24 of 24**. At the classical lens $d = 1/2$, where $\varphi \neq -\pi/2$, the same indices fall across all four residues.

Under the null hypothesis of no phase structure the probability is $4^{-24} \approx 3 \times 10^{-15}$.

4. The far end, and Theorem 3

4.1 Why a large lens hides

[P] With $z = \beta + iT$, $q_a(\rho) = -(1 + z/d)/(1 - z/d) = -\exp(2z/d + O(d^{-3}))$, so

$$r(d) = 2\beta/d + O(d^{-3}), \quad \varphi(d) = -\pi + 2T/d + O(d^{-3}),$$

and $q_a(\rho) \rightarrow -1$. Hence $C_n \rightarrow 4(1 - (-1)^n)$: 0 for even n , 8 for odd. The next order decides the sign:

$$\begin{array}{ll} n \text{ even:} & C_n = 8n^2(T^2 - \beta^2)/d^2 + O(d^{-4}) \quad \rightarrow \text{positive when } T > |\beta| \\ n \text{ odd:} & C_n = 8 + 8n^2(\beta^2 - T^2)/d^2 + O(d^{-4}) \quad \rightarrow 8 \end{array}$$

[M] Ratios of exact to model at $n = 2, 4, 10, 50$: at $d = 2 \times 10^5$ they are 1.0000 throughout; at $d = 2 \times 10^4$ they are 1.0000, 1.0000, 0.9999, 0.9981 — the model is an expansion in nd^{-1} , so the departure grows with n and shrinks with d , which is what the two columns show. At $d = 10^7$: $C_4 = 1.15 \times 10^{-9}$, $C_5 = 8.000000$.

So for $T > |\beta|$ every C_n is positive once d is large enough, and since on-line zeros contribute $2[1 - \cos n\theta] \geq 0$, the whole coefficient is non-negative for the first m orders. That is the mechanism of Theorem 3.

4.2 The order of the quantifiers

[C] Fix d and let $n \rightarrow \infty$: $|q_a(\rho)| > 1$ whenever $\beta > 0$, so the violation appears. Fix n and let $d \rightarrow \infty$: the contribution is non-negative. The limits do not commute, and the statement is

$$\forall m \exists d_m, \quad \text{not} \quad \exists d \forall m.$$

Masking is temporary. No finite shift hides a zero forever, which is what the criterion's equivalence to RH requires.

4.3 The cost, measured

[M] With 2000 ordinates and the quartet at $T = 30$:

d/R	1	2	10	50	200
$N(d)$	2276	2863	11536	56947	227462
$N(d)/(d/R)$	2276	1432	1154	1139	1137

The index grows linearly in the lens width, the ratio settling by $d = 50R$. So pushing detection past order m costs a shift proportional to m , which is the scale $O(mR)$.

4.4 The U, and the scaling law

[M] Over $d/R = 0.005$ to 20 at $M = 2000$:

d/R	0.005	0.1	0.6	1	2	5	20
$N(d)$	229947	11568	2581	2276	2863	5936	22857

a clean U with its floor at $d = R$. And at the matched lens, over ten values of M from 250 to 100,000:

M	250	1000	4000	16000	64000	100000
$N_M(R)$	1660	2072	2500	2880	3296	3520
$N \cdot r - \log M$	+0.012	-0.001	+0.039	-0.080	-0.080	+0.220

$$N_M(d) \cdot r(d) = \log M + O(1).$$

[C] The $O(1)$ is not $o(1)$. The residual stays bounded — the full range over the ten values is $[-0.080, +0.220]$ — but it does not settle toward zero, and the last point is three times any earlier one. The law is an envelope with a bounded, non-vanishing correction, and should not be written with $o(1)$.

5. The symmetry, in the index

Proposition 2 is an exact statement about the rate. The detection index also depends on the background, which has no reason to respect $d \mapsto R^2/d$.

[M] It nearly does. Comparing $N(0.1R)$ with $N(10R)$:

M	2000	4000	8000	16000	32000	48000	64000	100000
ratio	1.0028	0.9936	0.9965	0.9871	1.0029	1.0212	1.0291	1.0217

A lens k times too narrow and one k times too wide cost within a few percent of the same number of coefficients, across a 50-fold range of M .

[C] But the departure is not noise at large M . The sign is positive at $M = 2000$, negative at 4000, 8000 and 16000, and then positive at **every** value from $M = 32000$ on — four consecutive values, reaching the 2 – 3% level from $M = 48000$: the narrow lens costs *more*. A plausible reading is that the two lenses interact differently with the truncation of the ordinate list, since at $d = 0.1R$ the topmost ordinates are still phase-coherent while at $d = 10R$ they have long since begun to oscillate.

[O] That reading is not supported by the data as it stands. The natural candidate variable, $\Lambda = 2N(d)d/\gamma_M$, falls monotonically from 27.6 to 1.4 across the range at $d = 0.1R$ and never crosses 1, while the sign changes twice. A monotone quantity cannot produce a non-monotone sign, so whatever fixes the sign is not Λ alone.

6. Two things that do not work

Both look like they should, which is the reason for recording them.

6.1 Phase-tuned lenses do not improve detection

Since $r'(R) = 0$ while $\varphi'(R) = -1/T$, a small displacement changes the phase to first order and the rate only to second. It is therefore tempting to displace the lens by $O(T/n)$ so that $\cos(n\varphi) = 1$ at a chosen order, losing only $O(n^{-2})$ of the rate. For a *fixed* n that is correct.

[M] It does not help the first index. Scanning d over $[R - 3, R + 3]$ with 2000 ordinates at $T = 30$, the eight displaced lenses at $d - R = -3, -1.5, -0.75, -0.25, +0.25, +0.75, +1.5, +3$ give, in that order, 2298, 2295, 2289, 2300, 2276, 2284, 2289, 2293, against $N = 2276$ at $d = R$ itself. **None is below.** One — at $d - R = +0.25$ — ties, so the minimum is attained twice and never beaten; every other lens costs between 8 and 24 coefficients.

Why. Minimising C_n at a fixed n is not the problem. The detection index is a **first passage over all** n , and displacing the lens buys alignment at one order while costing envelope at every order. The refinement also needs n in advance, which is what a search does not have.

6.2 The far detection pockets are swamped

[P] At large d , $\cos(n\varphi) = (-1)^n \cos(2nT/d)$, so for even n the phase realigns whenever $2nT/d = 2\pi k$. This gives a family of narrow **detection pockets** at

$$d \approx nT/(\pi k), \quad k = 1, 2, 3, \dots$$

of relative full width $2\beta/T$. **[M]** The measured centre approaches the prediction as n grows — relative departure 2.1×10^{-3} , 3.4×10^{-4} , 3.0×10^{-5} at $n = 40, 100, 400$ — because the minimum of C_n sits where the cosh and the cos balance, not exactly where the phase realigns; and the width is 0.00667, 0.00667, 0.00666 at $(n, k) = (100, 1), (400, 1), (400, 2)$ against $2\beta/T = 0.00667$.

[M] Not one of them produces a negative coefficient.

n, k	40, 1	100, 1	400, 1	400, 2
C_n in the pocket	-8.70×10^{-4}	-8.76×10^{-4}	-8.77×10^{-4}	-3.51×10^{-3}
background	3744.3029	3983.2184	3989.5495	3988.6781
B_n				
$S_n = B_n + C_n$	3744.3021	3983.2175	3989.5487	3988.6746

Why they never deepen. Along a pocket line the quartet's contribution is $C_n \approx -8n^2\beta^2/d^2 = -8(\pi k\beta/T)^2$, which **does not depend on n at all**. However far out one goes and however large the order, the pocket has the same depth — about 8.8×10^{-4} here — against a background of order the number of ordinates. The ratio is 2×10^{-7} .

A reader of the far-field analysis would reasonably expect a second family of usable lenses. There is none.

7. What this changes, and what it does not

It changes how a numerical search should be reported. A search that computes m coefficients, finds them all positive, and concludes that no violation is present at that height has established nothing unless it also states its shift. Theorem 3 says a large enough shift makes the first m positive regardless, §4.3 measures the cost as linear in the shift with threshold $O(mR)$, and §3.2 says that even at the optimal lens only one order in four carries the signal. **Three separate reasons why a finite check can come back clean and mean nothing.**

It does not establish that the optimal shift is worth choosing. Sekatskii's own position (correspondence, August 2026) is that the maximum of the modulus, and hence the choice of a , **might** be rather than is important for numerical work, and that qualification is adopted here. The masking direction is the firmer of the two: it rests on his Theorem 3, which is proved.

It does not touch the Riemann Hypothesis, and it does not make the criterion easier to prove. Every statement here is about a detector's behaviour on a planted configuration, in a finite truncation.

Knowing when an instrument is blind is not the same as improving it.

8. Reproducibility

```
ZETA_ZEROs=zeros.txt python code/verify_phase_masking.py          # 13 checks, ~85 s
ZETA_ZEROs=zeros.txt python code/verify_phase_masking.py --fast    # ~3 s
```

Every number above comes from that script, which exits nonzero if any check that could be run fails.

ZETA_ZEROs names a file of critical-line ordinates, one per line, increasing. The measurements above use the first **100,000**, γ from 14.1347 to 74920.8275; a list of that size can be taken from Odlyzko's tables (www.dtc.umn.edu/~odlyzko/zeta_tables) or the LMFDB, or generated with *mpmath.zetazero*, which is far slower. Without the variable the symbolic and quartet checks still run and the sweeps are skipped.

A shorter list does not silently become a longer one. numpy slicing is quiet — *gam*[: 100000] on a 2000-element list returns 2000 — so every slice in the script is guarded, and a table needing more ordinates than the list holds is skipped and listed under *REDUCEDRUN* at the end rather than computed on fewer and printed under the count it asked for. A short list is not a failure and does not set the exit code; it is reported so that no printed number carries a label it did not earn. In particular §5 needs $M = 32000$ before it can say anything about the sign, and §4.4's reading of the residual needs the full ten-point run.

Tested with Python 3.12, NumPy 2.4, SymPy 1.14.

AI assistance

The verification script and much of the prose in this note were written with the assistance of Claude (Anthropic); the research direction, the decisions about what to publish, and responsibility for every claim are the author's. See the repository README for a fuller statement.

Acknowledgements

I am grateful to **S. K. Sekatskii** for correspondence on the relation between the measurements here and his Theorem 3, and in particular for confirming that the theorem is obtained through the analytic formula for ξ'/ξ and so is a statement about ξ , not about general multisets. Any error that remains is mine.

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No claim is made about the Riemann Hypothesis. Everything here is a measurement of a detector's behaviour on a planted configuration in a finite truncation, not evidence about the location of any zero.